



Dr. Yellapragada Subba Rao,
Wonder-man of **Miracle Drugs**

Unit

6

Biology in Human Welfare

THE '**UNINVITED GUESTS**' AND THE '**SILENT EPIDEMICS**' THEY CAUSE

- A PEEP INTO THE WORLD OF **PARASITES** vis-à-vis **MAN**

At least one-third of all animal species are estimated to be parasites according to one estimation. In ancient Greece, a person, who flattered and amused the host in return for free meals (a **professional dinner guest**), was used to be called a parasite.

It is stated that the single most '**undiagnosed health challenge**' in the **history of the human race** is '**parasites**'. More than 130 types of parasites are known to invade humans as hosts. As many people are not aware of the presence of certain parasites until it is too late to be helped, parasitism is described as a '**Silent Epidemic**'. **Most parasites tend to disturb the immunological system.**

Diseases such as **malaria, polio, plague, amoebiasis, dengue, filariasis, taeniasis**, are more common in the **tropical countries**. Due to the efforts of WHO, Small Pox was wiped out decades ago. Plague, which killed millions in the past, is almost a **part of history now**. However malaria is still evading control. Education on personal hygiene, providing proper sanitation facilities by the local administrations, will go a long way in preventing many tropical diseases.

Did the parasite come first or the host? It is believed that certain organisms enter the body of another organism accidentally and settle in it, slowly turning into a parasite. Have you ever heard that in the course of evolution, certain **prokaryotic organisms** entered the bodies of **eukaryotes** and remained in them for ever, acting as integral parts of the eukaryote's cells, and even playing a vital role in the life of

the eukaryote. **Can you guess one such prokaryote?** Did it ever occur to you that **mitochondria** could be one such type of prokaryotes?

On the other hand, the youth are getting addicted to **drugs** and **alcohol** – an influence of the western culture, spread by the television and movies. The harmful effects of **tobacco** are well publicised nowadays and the youth are sufficiently educated on that.

WONDER-MAN OF MIRACLE DRUGS

Probably many of you might not have heard the name of **Dr. Yellapragada Subba Rao**, ‘the greatest Indian Biochemist’ who made a dynamic impact in the field of Biochemistry and Medicine with a large number of discoveries which transformed the medical science and saved millions of people from deadly diseases.

Dr. Yellapragada Subba Rao was born in a poor family at Bhimavaram in West Godavari district of Andhra Pradesh. While he was studying in Madras Medical College, his elder brother and younger brother died of ‘**Tropical sprue**’ disease in the span of 8 days. He was deeply affected and decided to invent a medicine to cure ‘Sprue’. So he went to USA and took admission in the ‘Harvard School of Tropical Medicine’. Later he joined as Director in ‘Lederle Research Laboratory’ (now PFIZER), USA and continued there till his last breath.

His contributions in Biochemistry and Medicine

His contributions include: The discovery of **ATP** and **phosphocreatine** as the sources of energy in muscle contraction, **folic acid**, which is used in the treatment of **tropical sprue** that killed his two brothers, **anti-cancer** drug **Methotrexate**, which is also used in the treatment of non-cancerous diseases such as **rheumatoid arthritis** and **psoriasis**, **diethyl carbamazine-DEC** (popular brand name-**hetrazan**), the only effective drug in treating **filariasis**, the world’s first **Tetracycline** antibiotic **Aureomycin** which is the only drug effective in treating **cholera**, **plague**, **typhus fever**, **trench fever**, etc., and **isonicotinic acid hydrazide**, an effective medicine in treating **tuberculosis**.

Biology in Human welfare: Introduction

Biology is the youngest of the Natural Sciences. Progress in Physics and Chemistry proceeded much faster than in Biology. However, the 20th century and certainly the 21st century have demonstrated the utility of biological knowledge in furthering human welfare, medical field, agriculture and animal husbandry. The discovery of antibiotics, synthetic or plant-derived drugs, anaesthetics, diagnostic procedures, etc. have changed the medical practice on one hand and the human health on the other hand. Life expectancy of human beings has dramatically changed over the years due to the advancement in Biology. Biology also helps us to cope up with the changes that take place in our body due to various types of parasitic, bacterial, viral, fungal and other infections. It also educates people on use/abuse of tobacco, drugs, alcohol etc. These are briefly described in the following chapters of this unit.

- 6.1. Parasitism and parasitic adaptations
- 6.2. Health and disease
- 6.3. Brief account of some other diseases
- 6.4. Tobacco, Drugs and Alcohol abuse

6.1 PARASITISM AND PARASITIC ADAPTATIONS

Every living organism in this universe has an instinct to survive, for which, they require two basic needs - food and shelter. You know pretty well that green plants can prepare their own food materials in their body through photosynthesis, using CO_2 and H_2O as raw materials (**autotrophs**). But animals cannot prepare their own food materials in their body. Hence they must depend on plants or other animals for their survival (**heterotrophs**).

Certain animals are described as parasites as they survive at the cost of the other animals, called hosts. In such an association, the hosts tend to reject or resist the parasites. Concurrently the parasites have to evolve mechanisms to counteract and neutralize the host's defence in order to be successful within the host. For this purpose, the parasites have developed many special adaptations such as the loss of unnecessary sensory organs, formation of organs for adhesion, high reproductive capacity, etc.

6.1.1 Parasitism

An intimate association between two organisms of different species in which, 'one is benefited and the other one is often adversely affected' is called **parasitism**. The word parasitism comes from a Greek word '*parasitos*' (*para* - at the side of; *sitos* - food or grain) which means '*one eating at another one's table*'. The organism that obtains nourishment is called the '**parasite**' (the gainer) and the organism from which the nourishment is obtained is called the '**host**' (the loser).

I. Types of parasites

Based on the interaction between the host and the parasite, various types of parasites are recognised. Some of them are listed below.

- i) **Ectoparasite:** A parasite that lives on the surface of the host's body is called ectoparasite.

e.g. head lice and itch mites on humans, ticks on dogs, copepods on marine fishes, etc.

The female mosquito is not considered an obligate parasite though it needs human blood for fertility. Can you explain why?

Discuss with your teacher about the food habits of mosquitoes.

- ii) **Endoparasite:** A parasite that lives inside the body of the host is called endoparasite. Based on the place where they live, they are again classified into three types, namely:

- a) **Cytozoic (intracellular) parasites:** They live within the host's cells.

e.g. *Plasmodium vivax* in man, *Nosema notabilis* in *Sphaerospora polymorpha*, etc.

- b) **Histozoic (intercellular) parasites:** They live in between the cells of the tissues and organs of the host.

e.g. *Wuchereria bancrofti*, *Entamoeba histolytica*, etc.

- c) **Coelozoic parasites:** They live within the cavities of the host's body. They are called enterozoic if they live in the alimentary canal.

e.g. *Ascaris lumbricoides*

- iii) **Hyperparasite (Parasite in/on a parasite):** It is a parasite which lives in/on the body of another parasite.

e.g. *Nosema notabilis* (a cnidosporan parasite) lives in *Sphaerospora polymorpha* (also a cnidosporan parasite) which lives in the urinary bladder of toad fish.

- iv) **Monogenetic parasite:** It is a parasite which completes its life cycle in only one host.

e.g. *Entamoeba histolytica*, *Ascaris lumbricoides*, etc.

- v) **Digenetic parasite:** It is a parasite which requires at least two hosts to complete its life cycle.

e.g. *Plasmodium vivax*, *Wuchereria bancrofti*, etc.

NOTE: Parasites living in the gut of the host are called 'enterozoic parasites'. *Ascaris* is **enterozoic** and **coelozoic**, where-as *Entamoeba* is **enterozoic** and **histozoic**.

II. Types of Hosts

- i) **Primary host or Definitive host:** It is the host that harbours the adult stage or sexually mature stage of a parasite or the host in which the parasite undergoes sexual reproduction.

e.g. Man for *Wuchereria bancrofti*, female *Anopheles* for *Plasmodium*, etc.

- ii) **Intermediate host or Secondary host:** It is the host that harbours the developing 'larval or immature or asexual' stages of a parasite or the host in which the parasite undergoes asexual reproduction.

e.g. Man for *Plasmodium*, female *Culex* for *Wuchereria*, etc.

iii) Reservoir host: It is the host that lodges the infective stages of a parasite in its body when the human host is not available. In the reservoir host, the parasite neither undergoes development nor causes any disease.

e.g. Monkey for *Plasmodium*, African antelope (*Gnu*) for *Trypanosoma gambiense*, etc.

III. Vector

It is an organism (generally an insect) which transfers the infective stages of a parasite from one main host to another. Vectors are of two types, namely:

i) Mechanical vector: It is the vector, which merely transfers the infective stages of a parasite but no part of the parasitic development takes place in it.

e.g. House flies and cockroaches in the case of *Entamoeba*.

ii) Biological vector: It is the vector in which the parasite undergoes a part of the development before it gets transferred to another host.

e.g. Female *Anopheles* mosquito in the case of *Plasmodium* and female *Culex* mosquito in the case of *Wuchereria*.

6.1.2 Parasitic adaptations

Parasites have evolved special adaptations to meet the requirements and lead successful life in the hosts.

- ❖ In order to live in the host, some parasites have developed structures like hooks, suckers, rostellum, etc., for '**anchoring**'. **e.g.** *Taenia solium*
- ❖ Some intestinal parasites have developed **protective cuticle** to withstand the action of the digestive enzymes of the host. **e.g.** *Ascaris lumbricoides*
- ❖ Some intestinal parasites produce '**anti enzymes**' to neutralize the effect of host's digestive enzymes. **e.g.** *Taenia solium*
- ❖ Some parasites live as '**obligatory anaerobes**' as the availability of oxygen is very rare for them. **e.g.** *Entamoeba histolytica*, *Taenia solium*, etc.
- ❖ Some intestinal parasites live as '**facultative anaerobes**' i.e., if oxygen is not available, they live anaerobically and if oxygen is available, they respire aerobically. **e.g.** *Ascaris lumbricoides*
- ❖ The morphological and anatomical features are greatly simplified while emphasizing their **reproductive potential**. For example, an *Ascaris* lays nearly two lakh eggs per day. In *Taenia solium* the body is divided into 700 to 900 proglottids of which each proglottid acts as a unit of reproductive system and releases approximately 35,000 eggs.
- ❖ The life cycles of endoparasites are more complex because of their extreme specialization. For example, life cycle of certain parasites like *Fasciola hepatica* (sheep liver fluke) is very complex involving many developmental stages and

intermediate hosts, to increase the chances of reaching a new definitive host.

- ❖ Certain parasites like *Entamoeba* develop cysts to tide over the unfavourable conditions like desiccation while reaching the new host.
- ❖ Some parasites elude production of vaccines against them (smart parasites!) as they keep changing their surface antigens form time to time.
e.g. *Plasmodium*, HIV, etc.

Do you know: In villages, the eggs of ducks are mixed with those of hen and allowed to be incubated by hen- a kind of human induced 'brood parasitism'!

6.1.3 Effects of parasites on hosts

In general, the parasites cause weakening of the body of their hosts by causing the deprivation of nutrients, fluids and metabolites as they compete with their hosts for the same. They may also cause pathological effects in their hosts such as

- ❖ **Parasitic castration:** Some parasites cause the degeneration of gonads of the host, making it sterile. This effect is called **parasitic castration**.
e.g. *Sacculina* (root headed barnacle, a crustacean) causes the degeneration of gonads in the crab *Carcinus maenas*.
- ❖ **Neoplasia:** Some cause an abnormal growth of the host cells in a tissue to form new structures. This effect is called **neoplasia** which leads to cancers.
e.g. Some viruses
- ❖ **Gigantism:** Some parasites cause an abnormal increase in the size of the host. This effect is called **gigantism**.
e.g. The larval stages of *Fasciola hepatica* cause gigantism in snail (an intermediate host)
- ❖ **Hyperplasia:** Some parasites cause increase in the **number of cells**. This effect is called **hyperplasia**.
e.g. *Fasciola hepatica* in the bile duct of sheep
- ❖ **Hypertrophy:** Some parasites cause an abnormal increase in the **volume/size** of the infected host cells. This effect is called **hypertrophy**.
e.g. RBC of man infected by *Plasmodium*
- ❖ Most of the parasites cause various types of diseases like
 - i) African sleeping sickness by *Trypanosoma gambiense*
 - ii) Delhi boils/Tashkent ulcers/Oriental sores by *Leishmania tropica*
 - iii) Kala azar/Dum dum fever/Visceral leishmaniasis by *Leishmania donovani*
 - iv) Malaria by *Plasmodium* sps
 - v) Elephantiasis by *Wuchereria bancrofti*.

'An ideal parasite should be able to thrive within the host without harming it'. Do you believe in it? Then why didn't Natural Selection lead to the evolution of such totally harmless parasites? Clue: Think of the phenomenon of 'Ecological balance in nature'

6.2 HEALTH AND DISEASE

6.2.1 Health

The term health is very frequently used by everybody. How do we define it? For a long time, **'health'** was considered a state of body and mind where there was a balance of certain 'humors' according to **'Good humor'** hypothesis. This is what early Greeks like Hippocrates as well as the Indian Ayurveda system of medicine asserted. According to this hypothesis, it was thought that persons with **'black bile'** belonged to **hot personality** and **would have fevers**. This idea was arrived at by 'a pure reflective thought'. The discovery of 'the circulation of blood by William Harvey using experimental method' and 'the demonstration of normal body temperature using thermometer in persons with black bile' disproved the 'Good humor' hypothesis.

In later years, biology stated that mind influences our immune system through neural system and endocrine system and that our immune system maintains our health. Hence, mind and mental status can affect our health. **Health** is a state of complete physical, mental and social well-being and not merely 'absence of any disease' or 'absence of physical fitness'.

Of course, our health may be affected by–

- i) **Genetic disorders** – deficiencies with which a child is born or the defects inherited by the child from its parents.
- ii) **Infections** – by either animal parasites or bacteria or virus or fungi.
- iii) **Life style** – type of food and water we take, timings of food intake, amount of rest and exercise we give to our bodies, habits such as smoking, drug and alcohol abuse, etc.

When people are healthy, they are more efficient at work. This increases productivity and brings economic prosperity. Health also increases longevity of people and reduces infant and maternal mortality. Balanced diet, personal hygiene and regular exercise are very important to maintain good health. Since times immemorial, yoga has been practiced to achieve physical and mental health. Awareness about diseases and their effect on different bodily functions, vaccination against infectious diseases, proper disposal of biological wastes, control of vectors and maintenance of hygienic food and clean water resources are necessary for achieving good health.

6.2.2 Disease

Any change from the normal state of health that causes discomfort or disability is called **disease**. When the functioning of one or more organs or systems of the body is adversely affected characterized by various signs and symptoms, we say that we are not healthy or we have a disease. Diseases can be broadly grouped into two types namely - infectious and non-infectious.

- i) **Infectious diseases:** The diseases which are easily transmitted from one person to another are called infectious diseases. These are caused by pathogens. These are very common and every one of us might have suffered from any of these diseases, at sometime or the other.
e.g. Amoebic dysentery, Malaria, Elephantiasis, Typhoid, Pneumonia, Common cold, Ringworm, etc.
- ii) **Non-infectious diseases:** The diseases which are not transmitted from one person to another and are not caused by pathogens are called non-infectious diseases.
e.g. Genetic disorders, kidney problems, heart problems, etc.

6.2.3 Common Parasites causing diseases in man

A wide range of forms that belong to protozoans, helminths, bacteria, fungi, viruses, etc., could cause diseases in man. Such disease causing forms are called pathogens. The pathogens can enter our body by various means, multiply and interfere with normal vital activities, resulting in morphological and physiological damage. A few representative members from different groups of pathogenic organisms are discussed here along with their life cycle, pathogenicity, treatment and preventive measures against the diseases caused by them.

6.2.3.1 *Entamoeba histolytica*

I. Systematic position

Phylum : Protozoa
 Subphylum : Sarcomastigophora
 Class : Rhizopodea.

Do you know the reason for the death of the famous Mughal emperor **Akbar** and the most famous Telugu King **Sri Krishna Devaraya**?

Entamoeba histolytica (Gr: *entos* - within; *amoibe* - change; *histos* - tissues; *lysis* - dissolve) is a microscopic and monogenetic parasite that inhabits the large intestine and causes **amoebic dysentery** or **amoebiasis** in man. The mode of infection is through contaminated food and drinking water. House flies, cockroaches, etc., act as 'mechanical vectors'.

It is cosmopolitan in distribution but more common in the tropical and subtropical regions of the world. It is common in the people of rural and densely populated urban areas wherever the hygienic conditions are poor.

II. Structure

Entamoeba histolytica passes through three distinct stages in its life cycle, namely:

- i) Trophozoite stage

- ii) Precystic stage and
- iii) Cystic stage

1. Trophozoite stage: It is the most active, motile, feeding and **pathogenic stage** that lives in the mucosa and sub-mucosa membranes of the large intestine. It moves with the help of a single blunt finger like pseudopodium called lobopodium which is produced in the advancing end. The body of the trophozoite is surrounded by plasmalemma. Its cytoplasm is differentiated into an outer clear, viscous, non-granular ectoplasm and the inner fluid like, granular endoplasm. Ribosomes, food vacuoles and a vesicular, **cartwheel shaped nucleus** are present in the endoplasm. However contractile vacuoles, endoplasmic reticulum, Golgi apparatus and mitochondria are absent. The absence of mitochondria indicates the '**obligate anaerobic nature**' of *Entamoeba histolytica*. It produces the proteolytic enzyme called histolysin due to which the species name '**histolytica**' was assigned to it. Due to the effect of this enzyme, the mucosa and sub-mucosa of the gut wall are dissolved releasing some amount of blood, tissue debris which are ingested by the trophozoites. Hence, the food vacuoles are with erythrocytes, fragments of epithelial cells and bacteria. The mode of nutrition is **holozoic**. Presence of '**RBC in food vacuoles**' and cartwheel shaped nucleus are the characteristic features of the trophozoites of *Entamoeba histolytica*.

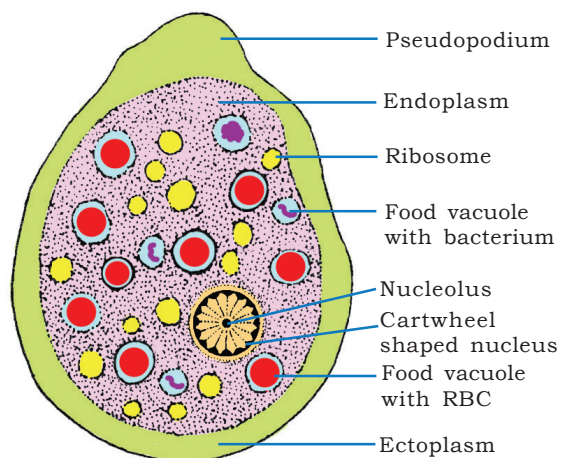


Fig. 6.1 Trophozoite Stage

2. Precystic stage: It is the non-feeding and non-pathogenic stage of *Entamoeba histolytica* that is found in the lumen of the large intestine. It is a small, spherical or oval, non-motile form. The cytoplasm of the precystic stage stores **glycogen granules** and **chromatoid bars** (made of ribonucleo protein) which act as reserve food.

3. Cystic stage: It is round in shape and is surrounded by a thin, delicate and highly resistant cyst wall. It is found in the lumen of the large intestine. The process of development of cyst wall is called **encystation** which is a means to tide over the unfavourable conditions that the parasite is going to encounter while passing to a new host. Soon after encystation, the nucleus undergoes two successive series of mitotic divisions to

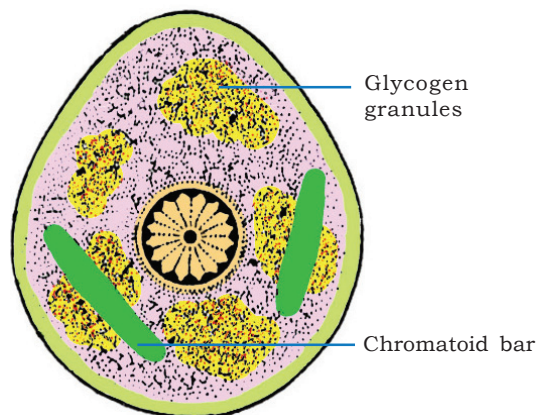


Fig. 6.2 Precystic stage

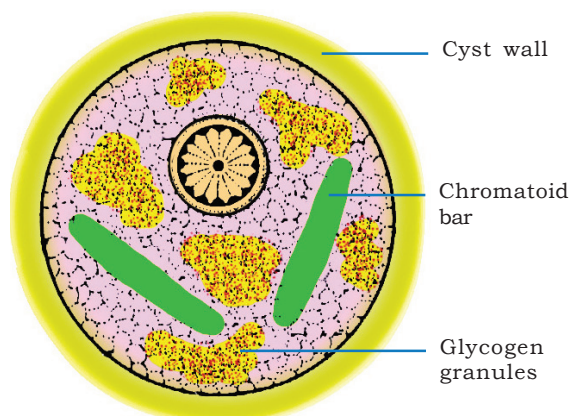


Fig. 6.3 Cystic stage

form **four daughter nuclei**. This type of cystic stage is called **tetra nucleate cyst** or **mature cyst** which is '**the stage infective to man**'.

III. Life cycle:

The trophozoites undergo binary fissions in the wall of the large intestine and produce a number of daughter entamoebae. They feed upon the bacteria and the host's tissue elements, grow in size and again multiply. After repeated binary fissions, when the trophozoites increase in number, some of the young ones enter the lumen of the large intestine and transform into **precystic stages**. Here, the precystic stages transform into **cystic stages** which

in turn develop into **tetranucleate cysts**. The entire process is completed only in a few hours. These tetranucleate cysts come out along with the faecal matter and can remain alive for about 10 days. These cysts reach new host through contaminated food and water. They pass into the small intestine of a new human host where the cyst wall gets ruptured by the action of the enzyme trypsin, releasing the **tetranucleate amoebae**. Such tetranucleate excystic amoebae are called **metacysts**.

The four nuclei of the metacyst undergo mitotic divisions and produce eight nuclei. Each nucleus gets a bit of the cytoplasm and thus eight daughter entamoebae or '**metacystic trophozoites**' are produced. These young ones develop into feeding stages called trophozoites. They invade the mucous membrane of the large intestine and grow into **mature trophozoites**.

IV. Pathogenicity

1. Intestinal amoebiasis

The trophozoites 'dissolve' the mucosal lining by **histolysin**, go deep into sub-mucosa and cause ulcers. These ulcers contain cellular debris, lymphocytes, red blood corpuscles and bacteria. It leads to the formation of abscesses in the wall of large intestine. Ultimately it results in stool with blood and mucous. This condition is called **amoebic dysentery** or **intestinal amoebiasis** or **tropical amoebiasis**.

2. Extra-intestinal amoebiasis

Some times, the trophozoites may rupture the wall of capillaries, enter the blood stream and primarily reach the liver where they may cause 'abscesses' (some call it 'secondary amoebiasis'). From there, they may go to lungs, heart, brain, kidneys, gonads, etc., and cause abscesses in those parts leading to **severe pathological conditions**.

3. Some people do not exhibit any symptoms. Such people are called '**carriers** or **asymptomatic cyst passers**' as their stool contains the tetranucleate cysts. They help in spreading the parasites to other persons.

In your view what could be the substance that forms the wall of the cyst in *Entamoeba histolytica*? Is it chitin/ protein/ lipid?

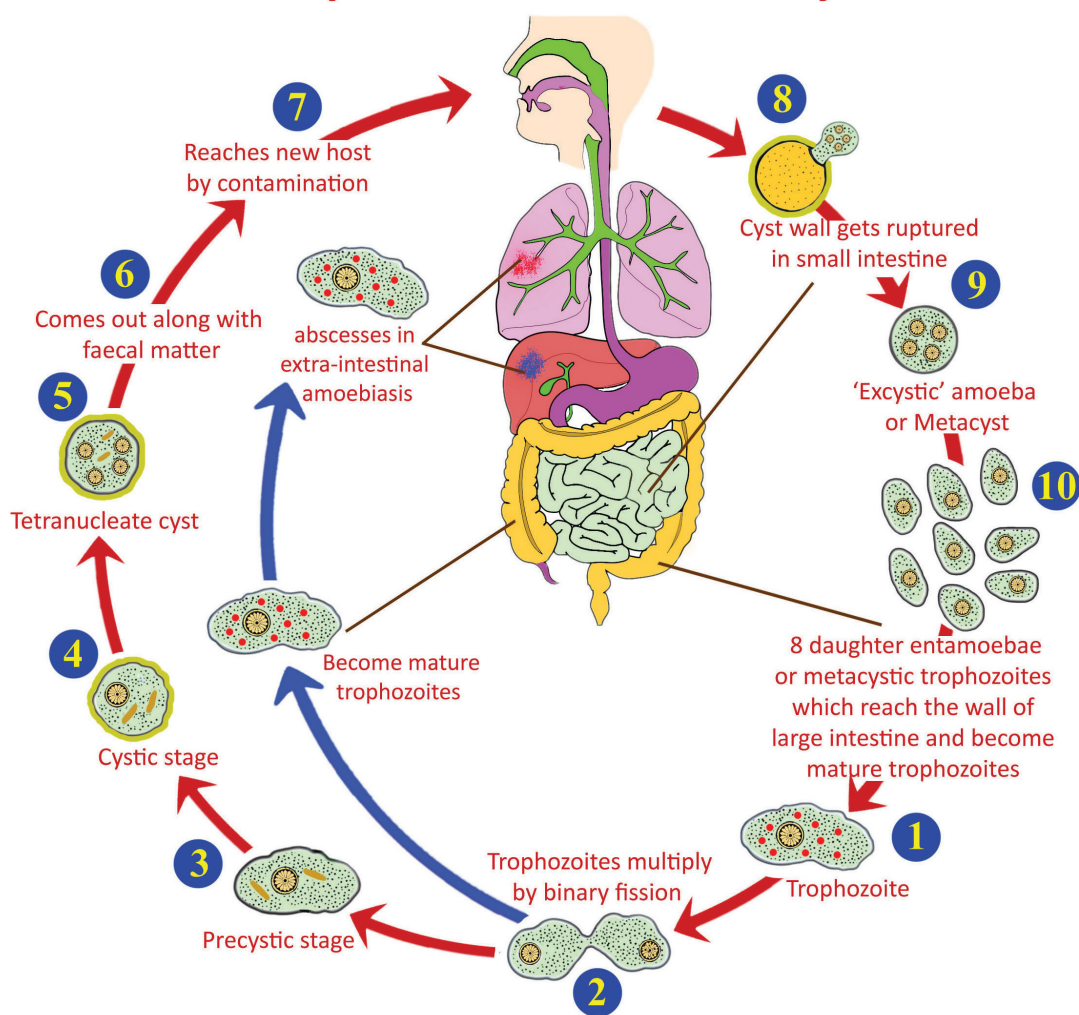


Fig. 6.4 Life cycle of *Entamoeba histolytica*

V. Prophylaxis

The following hygienic habits help prevent spread of this disease:

- Using boiled and filtered water
- Washing hands, fruits and vegetables properly
- Using septic tank toilets

In some cases, a permanent cure for amoebiasis is very hard to achieve as it frequently recurs unless the complete course of medication prescribed by a qualified medical practitioner is used.

6.2.3.2 *Plasmodium vivax*

Systematic position

Phylum	: Protozoa
Subphylum	: Apicomplexa (Sporozoa)
Class	: Telosporea
Genus	: <i>Plasmodium</i>
Species	: <i>vivax</i>

Do you know the reason for the death of '**Alexander the Great**'? No one was able to defeat **Alexander**, but the tiny Indian mosquitoes 'defeated' him. Do you know how?

Among the protozoans, *Plasmodium* is one of the most harmful parasites of man. It is a **digenetic, intracellular parasite** that lives in the **liver cells** and **RBC** of man. (It is **extracellular** in mosquito). Its **primary host** is the female **Anopheles** mosquito and the **secondary host** is man. Reservoir host is monkey. The infective stage is **sickle shaped sporozoite** and the mode of infection is **inoculation**.

Four species of *Plasmodium* cause four types of malaria in man. They are

- i) *Plasmodium vivax* – benign tertian malaria
- ii) *Plasmodium falciparum* – malignant tertian malaria
- iii) *Plasmodium ovale* – mild tertian malaria
- iv) *Plasmodium malariae* – quartan malaria

Of all these four species, *Plasmodium vivax* is the most common and most widely distributed malaria parasite. Hence its life cycle along with pathogenicity, treatment and preventive measures are briefly described here.

I. History of Malaria

The term '**Malaria**' is taken from Italian language which means 'bad air' (*Mala* - bad; *aria* - air) as it was thought that malaria was due to foul air. **Charles Laveran**, a French military doctor discovered *Plasmodium* in the blood of a malaria patient. **Sir Patrick Manson**, a Scottish doctor while working on elephantiasis, suggested that malaria might be spread by mosquitoes. **Sir Ronald Ross**, a British army doctor, while working in Secunderabad, A.P., India, identified the oocysts of *Plasmodium* in the stomach wall of female *Anopheles* mosquito on the 20th of August. In recognition of this work, Ross was awarded the **Nobel Prize** in 1902 and 20th August of every year is celebrated as **World Mosquito Day**. April 25th is the World Malaria day by WHO (from 2007). **G.B. Grassi** and others have described the life cycle of *Plasmodium vivax* in the female *Anopheles* mosquito.

II. Structure of sporozoite

The ultra-structure of the sporozoite of *Plasmodium vivax* was studied by **Garnham**. It is sickle shaped with a swollen middle part and pointed at both ends of its body. It measures about 15 microns in length and one micron in width. The body is covered by an elastic pellicle with '**microtubules**' which help in the wriggling movements of the sporozoite. The cytoplasm contains cell organelles such as Golgi complex, endoplasmic reticulum, mitochondria and a nucleus. Cytoplasm also shows many convoluted tubules of unknown function throughout the length of the body. It contains a cup like depression called **apical cup** at the anterior end into which a pair of **secretory organelles** opens. They secrete a cytolytic enzyme, which helps in the penetration of sporozoite into the liver cells.

The life cycle of *Plasmodium* is completed in two hosts as mentioned earlier.

III. Life cycle of *Plasmodium* in man (The human phase)

In man, the *Plasmodium* reproduces by asexual reproduction called **schizogony**. It occurs in liver cells (hepatocytes) as well as in RBC. In liver cells, it is called **hepatic schizogony** and in RBC it is called **erythrocytic schizogony**.

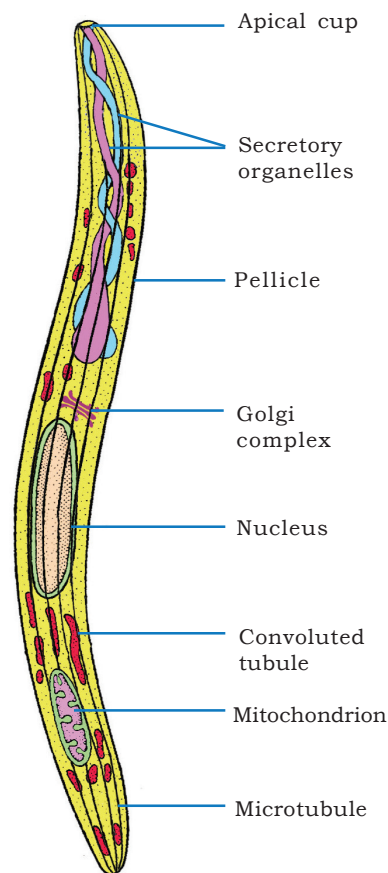


Fig. 6.5 Structure of sporozoite

1. Hepatic schizogony

This was discovered by **Shortt** and **Garnham**. Whenever, a mosquito infected by *Plasmodium* bites a man, nearly 2000 sporozoites are released into the blood of man through its saliva. Within half an hour, they reach the hepatocytes where they undergo **pre-erythrocytic** and **exo-erythrocytic** cycles.

A. Pre-erythrocytic cycle

Whenever the sporozoites reach the liver cells, they transform into **trophozoites**. They feed on the contents of the hepatic cells, assume spherical shape and attain the maximum size. This stage is called **schizont** stage. Its nucleus divides several times mitotically, followed by the cytoplasmic divisions resulting in approximately 12,000 daughter individuals called **cryptozoites** or the **1st generation merozoites**. They enter the sinusoids of the liver by rupturing the cell membrane of the schizont and the liver cells. This entire process is completed approximately in 8 days. Now these first generation merozoites have two options i.e. they can enter either fresh liver cells and continue exo-erythrocytic cycle or they can enter RBC and continue erythrocytic cycle.

Have you ever given a sample of blood for blood test? Did you observe how doctor takes the sample of your blood? Do you find any similarity between that and the bite of mosquito?

B. Exo-erythrocytic cycle

If the cryptozoites enter the fresh liver cells, they undergo the changes similar to that of the pre-erythrocytic cycle and produce the second generation merozoites called **metacryptozoites**. These are of two types- the smaller **micro-metacryptozoites** and larger **macro-metacryptozoites**. This entire process is completed approximately in two days. The macro-metacryptozoites attack fresh liver cells and continue another exo-erythrocytic cycle, whereas the micro-metacryptozoites always enter blood stream and attack fresh RBC to continue repeated erythrocytic cycles producing merozoites.

Prepatent period

The interval between 'the first entry of *Plasmodium* into the blood in the form of sporozoites and the second entry of *Plasmodium* into the blood in the form of cryptozoites is called **prepatent period**. It lasts approximately 8 days. During this period, the host does not show any clinical symptoms of the disease. It is only a means of multiplication.

2. Erythrocytic Schizogony

A. Cycle of Golgi

It was first described by **Camillo Golgi**. Hence it is also called '**Golgi cycle**'. This cycle is initiated either by the cryptozoites of pre-erythrocytic cycle or the micro-metacryptozoites of exo-erythrocytic cycle. In the fresh RBC, these stages assume spherical shape and transform into **trophozoites**. It develops a small vacuole which gradually enlarges in size, pushing the cytoplasm and nucleus to the periphery. Now the *Plasmodium* looks like a **finger ring**. Hence this stage is called **signet ring stage**. Soon it loses the vacuole, develops pseudopodia and becomes **amoeboid stage** (late trophozoite stage). With the help of pseudopodia, it actively feeds on the contents of the RBC and increases in size. As a result, the RBC grows almost double the size. This process is called **hypertrophy**. The malaria parasite digests the globin part of the ingested haemoglobin and converts the soluble haem into an insoluble crystalline **haemozoin**. It is called the '**malaria pigment**' which is a disposable product. During this stage, small red coloured dots appear in the cytoplasm of the RBC known as **Schuffner's dots**. These are believed to be the antigens released by the parasite. Now the *Plasmodium* loses the pseudopodia, increases further in size, occupies the entire RBC and becomes a schizont. It undergoes schizogony similar to that of the pre-erythrocytic cycle and produces 12 to 24 erythrocytic merozoites. They are arranged irregularly in the RBC form. Finally the erythrocyte bursts and releases the merozoites along with haemozoin into the blood. This cycle is completed approximately in 48 hours.

Incubation Period

The period between ‘the entry of *Plasmodium* into the blood in the form of sporozoite and the first appearance of symptoms of malaria in man’ is called **incubation period**. It is approximately 10 to 14 days.

B. Formation of gametocytes

After repeated cycles of erythrocytic schizogony, when the number of fresh RBC decreases, some merozoites enter the RBC and transform into gametocytes instead of continuing the erythrocytic cycle. This process generally takes place when the RBCs are present in **spleen** and **bone marrow**.

The gametocytes are of two types namely, smaller **microgametocytes** or **male gametocytes** and larger **macrogametocytes** or **female gametocytes**. The gametocytes cannot undergo further development in man as the temperature and pH of the blood of man are not suitable for further development. These gametocytes reach the blood circulation and wait to reach the next host. They degenerate and die if they are not transferred to mosquito within a week.

IV. Life cycle of *Plasmodium* in mosquito (The mosquito phase) – Ross cycle

When a female *Anopheles* mosquito bites and sucks the blood of a malaria patient, the gametocytes along with the other stages of the erythrocytic cycle reach the crop of mosquito. Here all the stages are digested except the gametocytes. Further part of the life cycle consists of:

- i) Gametogony
 - ii) Fertilization
 - iii) Formation of Ookinete & Oocysts
 - iv) Sporogony
- i) **Gametogony:** The formation of male and female gametes from the gametocytes is called **gametogony**. It occurs in the lumen of the crop of mosquito.

a. Formation of male gametes: During this process, the nucleus of microgametocyte divides into eight daughter nuclei called **pronuclei** which reach the periphery. The cytoplasm is pushed out in the form of eight flagella like processes. Into each flagellum like process, one pronucleus enters and forms a **micro gamete** or **male gamete**. These male gametes show lashing movements like those of flagella and get separated from the cytoplasm of microgametocyte. This process is called **exflagellation**.

b. Formation of female gamete: The female gametocyte undergoes a few changes and transforms into a female gamete. This process is called **maturation**. The nucleus of the female gamete moves towards the periphery and the cytoplasm at that point forms a projection. This projected region is called the **fertilization cone**.

Do you know: Why a minimum of ten days is required for the occurrence of the first symptoms of malaria in man after the entry in the form of sporozoites?

SEXUAL LIFE CYCLE IN MOSQUITO

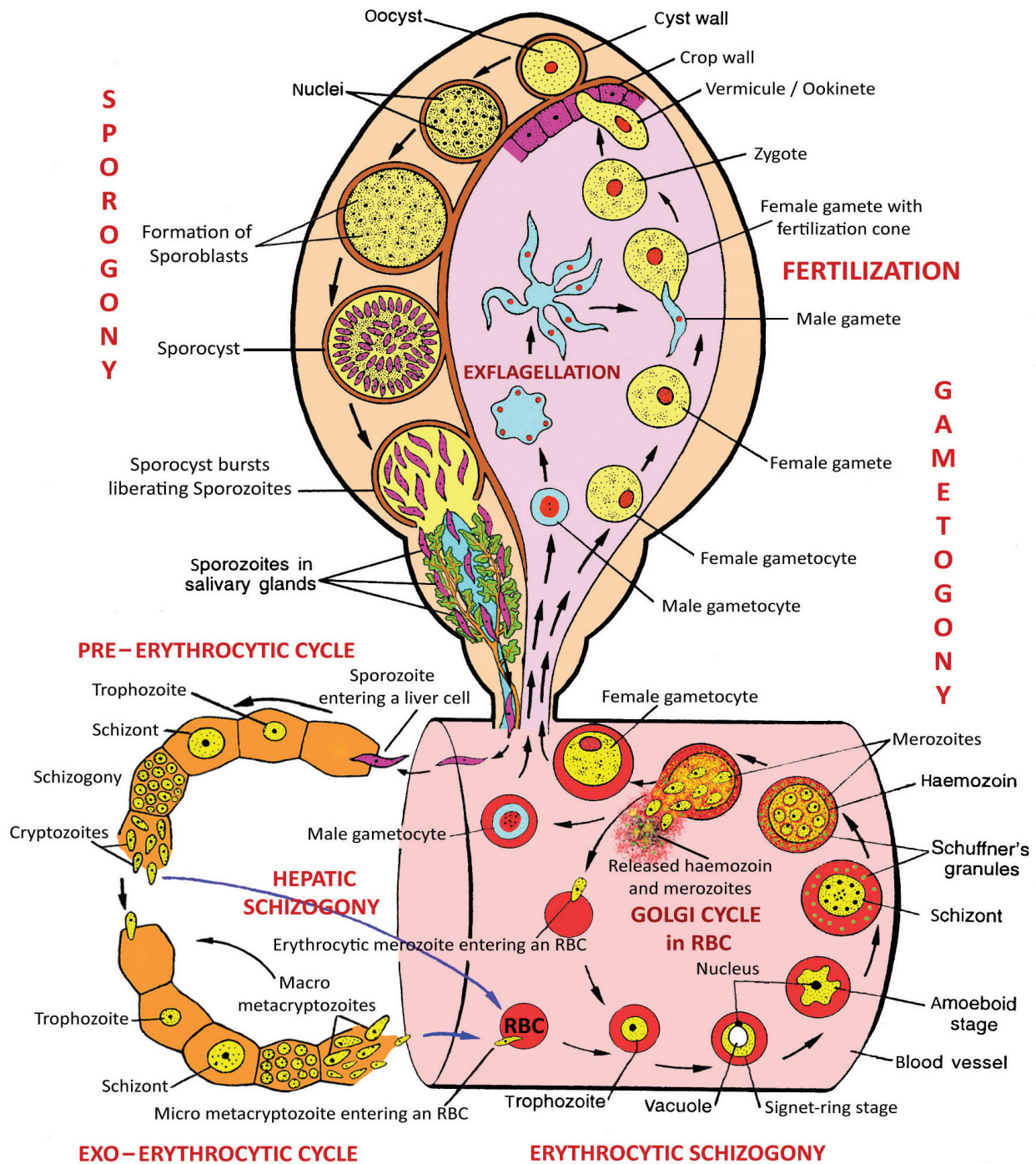


Fig. 6.6 Life cycle of *Plasmodium vivax*

- ii) **Fertilization:** The fusion of male and female gametes is called fertilization. It also occurs in the lumen of the crop of the mosquito. When an actively moving male gamete comes into contact with the fertilization cone of the female gamete, it enters it. The pronuclei and cytoplasm of these two gametes fuse with each other, resulting in the formation of a synkaryon. Since the two gametes are dissimilar in size, this process is known as **anisogamy**. The female gamete that bears the synkaryon is called the **zygote** which is round and non-motile.
- iii) **Formation of ookinete and oocysts:** The zygote remains inactive for some time and then transforms into a long, slender, motile, vermiform **ookinete** or **vermicule** within 18 to 24 hours. It pierces the wall of the crop and settles beneath the **basement membrane**. It becomes round and secretes a cyst around its body. This encysted ookinete is now called **oocyst**. About 50 to 500 oocysts are formed on the wall of the crop and appear in the form of small nodules. (Sir Ronald Ross identified these oocysts for the first time).
- iv) **Sporogony:** The formation of sporozoites in the oocysts is called **sporogony**. According to Bano, the nucleus of the oocyst first undergoes reduction division followed by repeated mitotic divisions resulting in the formation of sporoblasts. The nucleus of the sporoblast divides and forms several nuclei. Each nucleus is surrounded by a little bit of the cytoplasm and transforms into a sickle shaped sporozoite. Oocyst with such sporozoites is called **sporocyst**. When this sporocyst ruptures, about 10,000 sporozoites are liberated into the haemocoel of the mosquito. From there, they travel into the salivary glands and are ready for infection. The life cycle of *Plasmodium* in mosquito is completed in about 10 to 24 days.

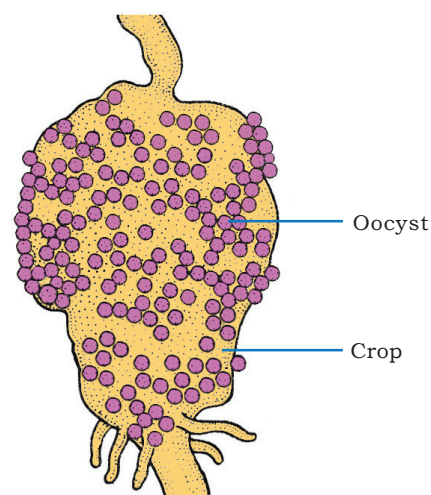


Fig. 6.7 Oocysts of *Plasmodium* on the crop of female *Anopheles*

Plasmodium exhibits alternation of generations in its life cycle. Can this statement be justified? How?

V. Pathogenicity

Plasmodium vivax causes benign tertian malaria. The clinical symptoms of this disease include bouts of fever which can be expressed in three stages namely cold stage, hot stage and sweating stage.

Cold stage: Chills, shivering, headache and giddiness.

Hot stage: High temperature of the body, rapid breathing and an increase in the pulse rate.

Sweating stage: Profuse sweating followed by lowering of the body temperature to the normal level.

Anaemia and **splenomegaly** are noticed in a chronic malaria patient. Relapse of malaria may also be noticed some times.

Why does the spleen enlarge in the case of chronic malaria patient?

Ans: In case of repeated infections, spleen shows an abnormal immune response and enlarges. (Tropical splenomegaly syndrome)

Relapse of malaria: Some of the stages of macro-metacryptozoites may survive for a long period in liver as dormant stages called **hypnozoites**. Reactivation of these hypnozoites leads to the initiation of fresh erythrocytic cycles resulting in the new attacks of malaria. This is referred to as **relapse of malaria**.

There are certain people who are immune to malaria. Do you know what the reason is for immunity against malaria ? (Ask your teacher to explain immunity to malaria in people heterozygous for sickle cell anaemia).

VI. Treatment

Malaria can be cured by **quinine** which is an alkaloid extracted from the bark of *Cinchona officinalis*.

VII. Prophylaxis

- ❖ Spraying of DDT, BHC, etc., insecticides at intervals in mosquito breeding places like pools, ponds, ditches and stagnant water.
- ❖ Introduction of larvivorous fishes like *Gambusia*, insectivorous plants like *Utricularia* into the places where mosquitoes breed.
- ❖ Avoiding mosquito bite by using mosquito nets, mosquito repellents, etc.
- ❖ Spraying of kerosene, pyrethrum oil, etc., on stagnant water.

6.2.3.3 *Ascaris lumbricoides*

Do you know the reason for giving the species name '*lumbricoides*' for *Ascaris*?

Clue : There is an earthworm called *Lumbricus*

Phylum : Nematoda

Class : Phasmidia

Ascaris lumbricoides is commonly called the **common round worm**. It is one of the most common entericparasites of man. It is monogenetic, dimorphic, pseudocoelomate parasite that lives in the small intestine of man, more frequently in children. It is cosmopolitan in distribution. Mode of infection is through contaminated food and water. Infective stage is the embryonated egg with the **2nd stage rhabditiform larva**.

I. Structure: Sexes are separate and the sexual dimorphism is distinct. In both males and females, the body is elongated and cylindrical. Mouth is present at the extreme anterior end and is surrounded by three chitinous lips. Close to the mouth mid ventrally, there is a small aperture called excretory pore.

1. Male: It has a curved posterior end which is considered the tail. The posterior end possesses a cloacal aperture and a pair of equal sized chitinous '**pineal spicules**' or '**pineal setae**' which serve to transfer the sperms during copulation.

2. Female: It has a straight posterior end, the tail. The female genital pore or vulva is present mid ventrally at about one third the length from mouth. The anus is present a little in front of the tail end.

II. Life history

Copulation takes place in the small intestine of man. After copulation, the female releases approximately two lakh eggs per day. Each egg is surrounded by '**a protein coat**' with rippled surface. Hence the eggs of *Ascaris* are described as '**mammillated eggs**'. The protein coat is followed by a chitinous shell and a lipid layer internally. These eggs are passed out along with faecal matter. In the moist soil, development takes place inside the egg leading to the formation of the **1st stage rhabditiform larva** is produced. It undergoes the **1st moulting** and becomes the **2nd stage rhabditiform larva** which is the stage '**infective to man**'. They reach the alimentary canal of man through contaminated food and water.

Do you know: 'If the conditions are favourable, the embryonated eggs of *Ascaris* can remain viable for about 6 years in moist soil'. Can you guess the reason for this?

In the small intestine, the shell gets dissolved and the 2nd stage larva is released. Now it undergoes **extra intestinal migration**. First it reaches the liver through the hepatic portal vein. From there it reaches the heart through the post caval vein. It goes to the lungs through the pulmonary arteries. In the alveoli of the lungs it undergoes the **2nd moulting** to produce the **3rd stage larva**. It undergoes the **3rd moulting** so that the **4th stage larva** is produced in the alveoli only. It leaves the alveoli and reaches the small intestine again, through the bronchi, trachea, larynx, glottis, pharynx, oesophagus and stomach. In the small intestine, it undergoes the

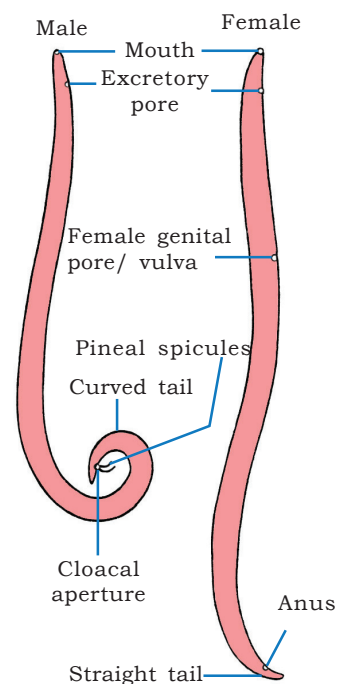


Fig. 6.8 *Ascaris lumbricoides*

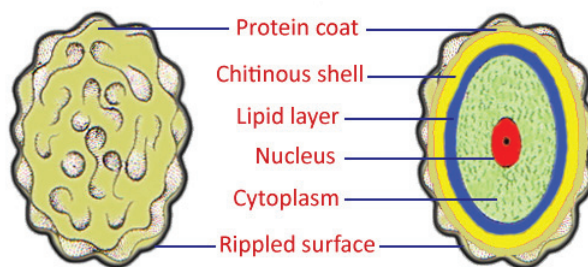


Fig. 6.9 Mammillated eggs of *Ascaris*

4th moulting (final moulting) to become a young one which attains sexual maturity within 8 to 10 weeks.

III. Pathogenicity

The disease caused by *Ascaris lumbricoides* is called '**ascariasis**'. The disease is asymptomatic if the number of worms is less. A heavy infection can cause nutritional deficiency and severe abdominal pain. It also causes stunted growth in children.

Try to find out the reason for stunted growth noticed in children due to *Ascaris lumbricoides* infection. (It is believed to produce certain substances which inhibit the Human Growth Hormone)

IV. Prophylaxis: Same as that of *Entamoeba*.

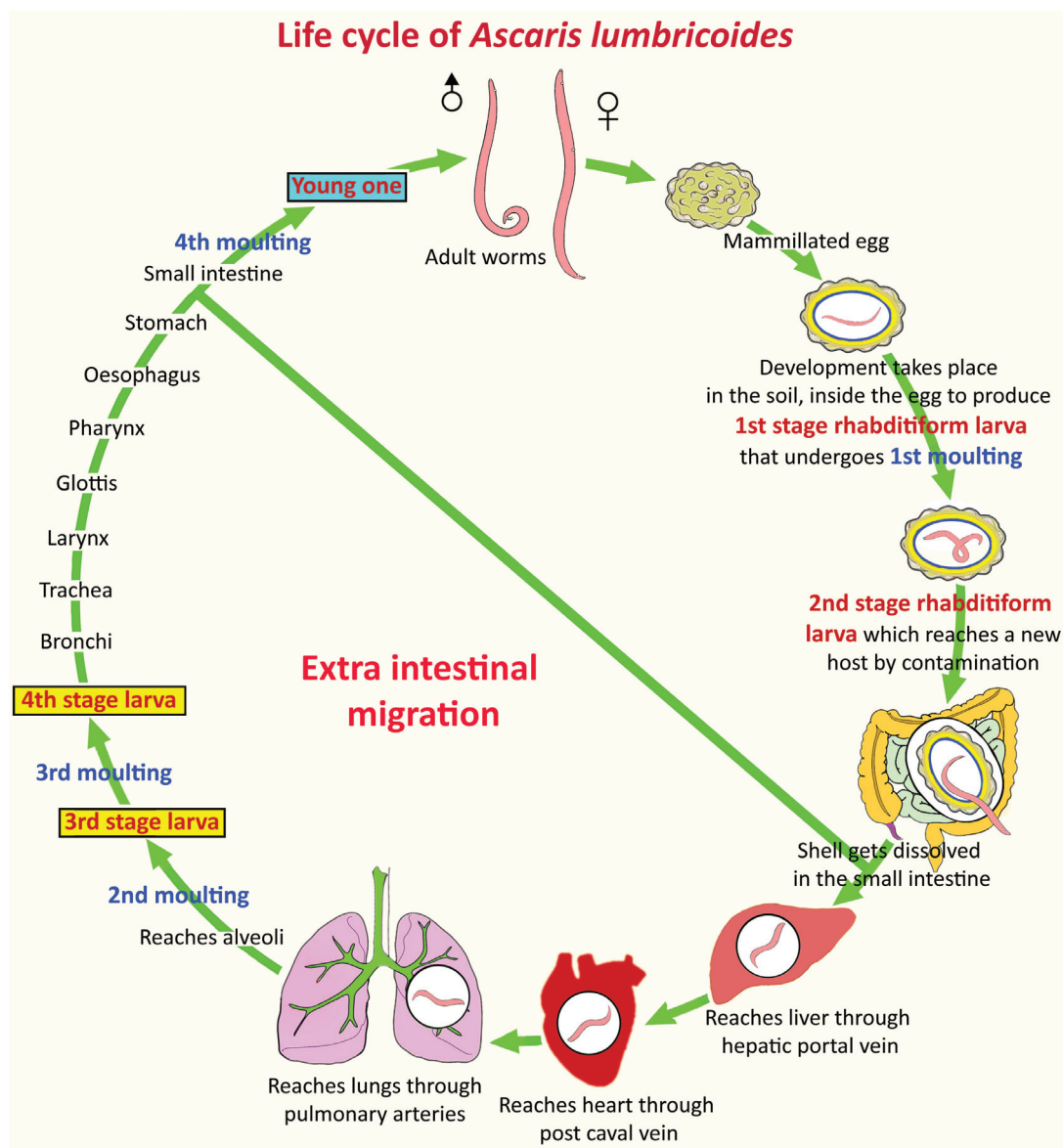


Fig. 6.10 Life cycle of *Ascaris lumbricoides*

6.2.3.4 *Wuchereria bancrofti*

Phylum : Nematoda

Class : Phasmdia

Wuchereria bancrofti is commonly called the filarial worm as it causes filariasis in human beings. It is a digenetic, dimorphic, pseudocoelomate and histozoic parasite that lives in the lymph vessels of man. **Sir Patrick Manson** identified **female Culex** mosquito as its secondary host.

I. Structure

Sexes are separate and the sexual dimorphism is distinct. The body is long, narrow, filiform and creamy-white in colour. The anterior end is blunt and the posterior end is pointed. Mouth is present at the anterior end and is without any lips.

1. Male worm: Its posterior end is curved with a cloacal aperture. A pair of unequal, chitinous '**pineal spicules**' or '**copulatory spicules**' is present in the cloacal region.

2. Female worm: Its posterior end is straight. Anus is present near the posterior end. The female genital pore or vulva is present mid ventrally at about one third the length from the mouth. It is **ovoviviparous**.

II. Life cycle

As we already discussed, it completes its life cycle in two hosts namely man and female *Culex* mosquito.

1. In man

Both male and female worms are found coiled together in the lymphatic vessels of man. After copulation the female releases the sheathed microfilaria larvae into the lymph of man. Each sheathed microfilaria larva measures 0.2 to 0.3 mm in length. It is surrounded by a loose cuticular sheath which is supposed to be the modified shell. They migrate to the blood circulation and reside in the deeper blood vessels during the day time. They move to the peripheral blood circulation during the night time between 10.00 pm and 4.00 am. This tendency is referred to as '**nocturnal periodicity**'. When a female *Culex* mosquito sucks the blood of an infected person, they enter the gut of mosquito. They die if they are not transferred to mosquito **within 70 days**.

Could you guess the reason for nocturnal periodicity of sheathed microfilaria?

2. In mosquito

In the midgut of mosquito, the sheath of the larva is dissolved within 2 to 6 hours of the infection. The ex-sheathed microfilaria larva penetrates the gut wall



Fig. 6.11 A case of elephantiasis

and reaches the haemocoel of mosquito. From there, it reaches the '**thoracic muscles**' and transforms into a '**sausage shaped larva**' within two days. It is called the **first stage larva** or **first stage microfilaria**. This undergoes two moultings within 10 to 20 days and transforms into a long, **infective '3rd stage microfilaria**'. It reaches the labium of the mosquito.

Why can't we consider the sheathed microfilaria larva the 1st stage microfilaria though it is the first stage produced in the life cycle of *Wuchereria*?

Note : A larva is a self supporting immature young one. Until the embryo can feed on its own, it can not be considered a larva.

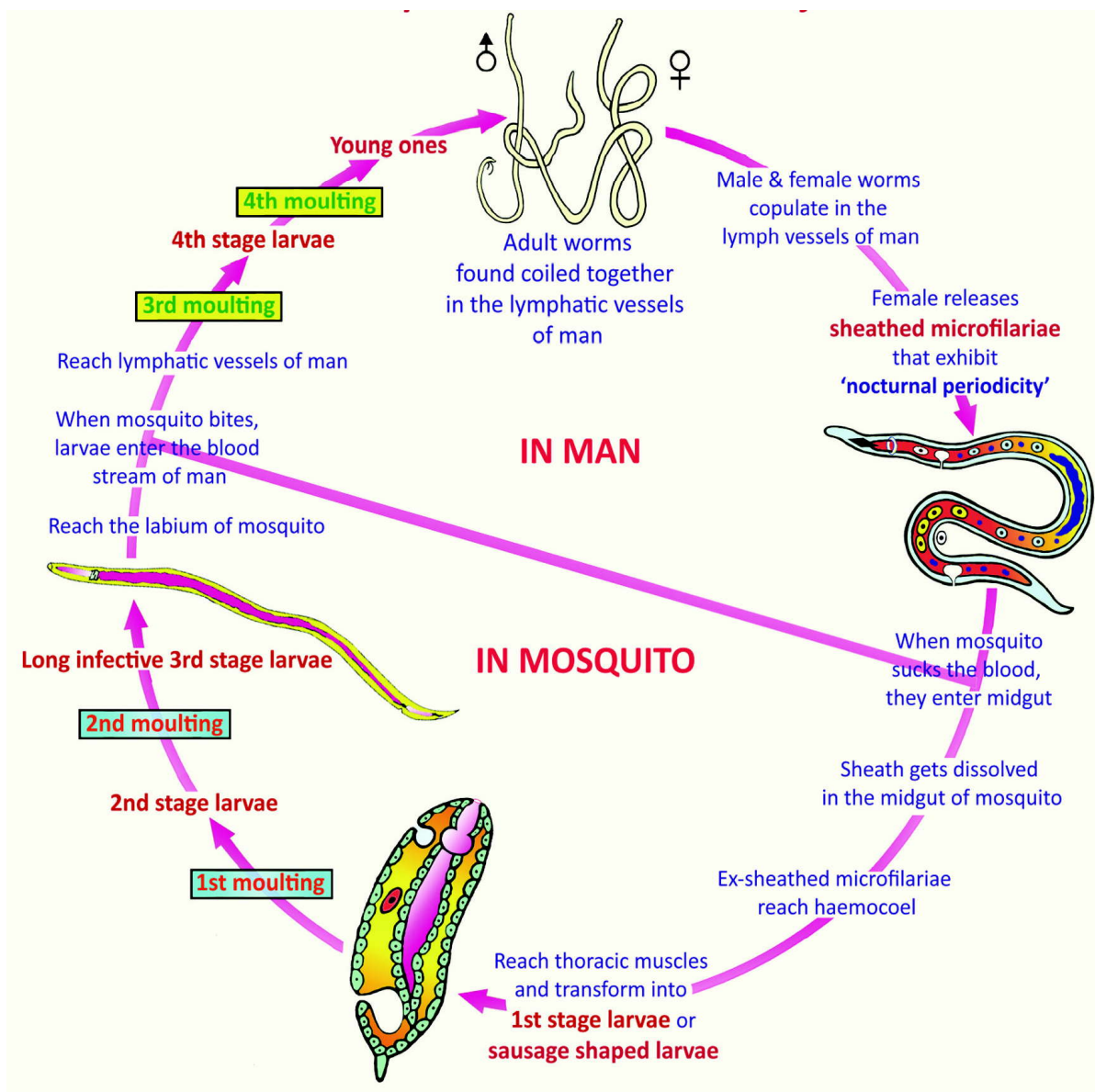


Fig. 6.12 Life cycle of *Wuchereria bancrofti*

3. In man after the infection

When an infected mosquito bites a man, the 3rd stage microfilaria larvae enter the blood circulation of man and finally reach the lymphatic vessels. Here they undergo the 3rd and the 4th moultings to produce young filarial worms. They attain sexual maturity within 5 to 18 months.

III. Pathogenicity

Light infection causes filarial fever which is characterised by headache, mental depression and increase in the body temperature. In general, the infection of filarial worm causes inflammatory effect in lymph vessels and lymph glands. Inflammation of the lymph vessels is called **lymphangitis** (Gr., *angeos* - vessels, *itis* - inflammation) and that of lymph glands is called **lymphadenitis** (Gr., *adenos* - gland, *itis* - inflammation). In the case of heavy infection, the accumulation of dead worms blocks the lymph vessels and lymph glands resulting in immense swelling. This condition is called **lymphoedema** (Gr., *oiedema* - a swelling) which is noticed in the extremities of limbs, scrotum of males and mammary glands in females. Fibroblasts accumulate in these tissues and form the fibrous tissue. In severe cases, the sweat glands of the skin in the affected regions disintegrate and the skin becomes rough. This terminal condition is referred to as '**elephantiasis**'.

IV. Prophylaxis: Same as that of *Plasmodium*.

6.3 BRIEF ACCOUNT OF SOME OTHER DISEASES

6.3.1 Bacterial diseases

- i) **Typhoid fever:** It is caused by *Salmonella typhi* which is a Gram negative bacterium. It mainly lives in the small intestine of man and then migrates to other organs through blood. It can be confirmed by **Widal test**.

Mode of infection: Contamination through food and water.

Symptoms: Sustained fever with high temperature upto 104°F, weakness, stomach pain, constipation, headache and loss of appetite. Intestinal perforation and death may also occur in severe cases.

A classic case in medicine: Mary Mallon nicknamed Typhoid Mary, was a cook by profession and was a typhoid carrier. She continued to spread this disease for several years through the food she prepared.

- ii) **Pneumonia:** It is caused by Gram positive bacteria such as ***Streptococcus pneumoniae*** and ***Haemophilus influenzae***. They infect the alveoli of lungs in human beings.

Mode of infection: Contamination by inhaling the droplets/aerosols released by an infected person or even by sharing the utensils with an infected person.

Symptoms: The alveoli get filled with fluid leading to severe problems in respiration. In severe cases, the lips and finger nails may turn gray to bluish in colour.

6.3.2 Viral Diseases

It is not an exaggeration to say that there is no person in this world who has not suffered from cold at least once in his life time.

Common cold: It is caused by **Rhino virus** group of viruses. They infect nose and respiratory passage but not lungs.

Mode of infection: Contamination by direct inhalation of the droplets resulting from cough or sneezes of an infected person or indirectly through contaminated objects such as pens, books, cups, door-knobs, computer keyboard or mouse etc.

Generally all the medicines that are used against cold cause drowsiness. Try to find out the reason for this from your family doctor.

Symptoms: Nasal congestion, discharge from nose, sore throat, hoarseness, cough, headache, tiredness, etc., which usually last for 3-7 days.

Do you know: 'Cold' treated is cured in just **ONE** week, and 'cold' untreated is cured in **SEVEN DAYS**. What is the implied meaning?

6.3.3 Fungal Diseases



Fig. 6.13 Ringworm infection

Ringworm: It is one of the most common infectious diseases in man. It is caused by many fungi belonging to the genera **Microsporum**, **Trichophyton** and **Epidermophyton**. Heat and moisture help these fungi grow in the skin folds such as those in the groin or between the toes.

Mode of infection: Contamination by using towels, clothes or combs of the infected persons or even from soil.

Symptoms: Appearance of dry, scaly, usually round lesions accompanied by intense itching on various parts of the body such as skin, nails and scalp.

6.3.4 Prophylaxis

- i) **In the case of bacterial & viral diseases:** The advancements made in biological science have armed us to deal with many infectious diseases effectively. The immunization programmes by the use of vaccines have enabled us to completely eradicate a deadly disease like smallpox. A large number of other infectious diseases like polio, diphtheria, pneumonia and tetanus have been controlled to a large extent by the use of vaccines.
- ii) **In general:** Biotechnology is at the verge of making available newer and safer vaccines. Discovery of antibiotics and various other drugs has also enabled us to treat infectious diseases effectively.

6.4 TOBACCO, DRUGS AND ALCOHOL ABUSE (TDA ABUSE)

Recent surveys and statistics show that the use of tobacco, drugs and alcohol has been on the rise especially among the youth. This is really a cause of concern as it could result in many harmful effects. Proper education and guidance would enable the youth to safeguard themselves against these dangerous habits and follow healthy lifestyles. Any addict requires counselling and medical help to get rid of the habit.

6.4.1 Tobacco

Tobacco has been used by human beings for more than 400 years. It contains a large number of chemical substances including **nicotine**, an alkaloid. While buying cigarettes one cannot miss the statutory warning printed on the packet '**Smoking is injurious to health**'.

Mode of abuse: It is smoked or chewed as 'gutkha' or used in the form of 'snuff'.

Effect: Smoking increases the carbon monoxide (CO) level and reduces the oxygen level in the blood. Nicotine stimulates the adrenal gland to release adrenaline and nor-adrenaline into blood. These hormones raise the blood pressure and increase the heart rate. Smoking is associated with bronchitis, emphysema, coronary heart disease, gastric ulcers and increases the incidence of cancers of throat, lungs, urinary bladder etc. Smoking also paves the way to addiction to hard drugs^[1]. Yet, smoking is very prevalent in society, both among the young and old. Tobacco chewing is associated with increased risk of cancer of the oral cavity.

Do you think **passive smoking** is also dangerous? Why?

6.4.2 Drugs

Drugs are the chemical substances used in the treatment, cure and prevention of diseases so as to enhance one's physical or mental well being. For hundreds of years, several plants, fruits and seeds with hallucinogenic properties have been used in folk-medicine, religious ceremonies (such as '**bhang**' on the '**Holi festival**' day) and rituals all over the globe. **When these are taken for a purpose other than the medicinal use or in excess amounts that impair one's physical or psychological functions, it constitutes 'drug abuse'.**

The drugs commonly abused are **opioids**, **cannabinoids** and **coca alkaloids**. Majority of them are obtained from flowering plants but some are obtained from certain fungi.

1. **Opioids:** These are the drugs obtained from opium poppy plant *Papaver somniferum* (vernacular name: '**Nallamandu**'



Fig. 6.14 Opium poppy

^[1] Hard drugs refer to morphine, heroin, cocaine, etc.

mokka'). They bind to specific opioid receptors present in our central nervous system and gastrointestinal tract. Some of the opioids are morphine, heroin, etc.

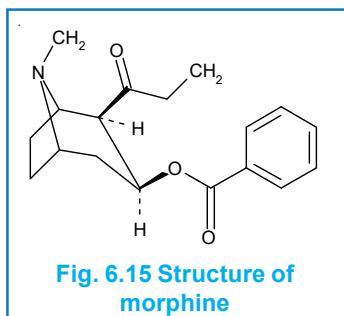


Fig. 6.16 Leaves of Cannabis

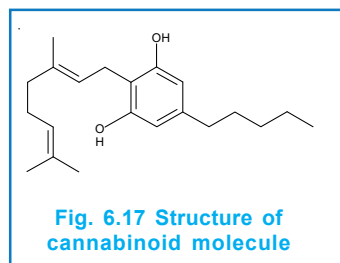


Fig. 6.18 A branch of Datura

- i) Morphine:** It is extracted from the dried latex of the unripe seed capsule (pod) of poppy plant. It occurs as colourless crystals or a white crystalline powder.

Mode of abuse: Generally it is taken orally or by injection.

Effect: It is a very effective sedative and painkiller. It is very useful in patients who have undergone surgery.

- ii) Heroin:** It is a white, bitter, odourless and crystalline compound, obtained by the **acetylation of morphine**. Chemically it is **diacetylmorphine**. It is commonly called '**smack**'.

Mode of abuse: Generally it is taken by '**snorting**' and injection.

Effect: Heroin is a depressant and slows down the body functions.

- 2. Cannabinoids:** These are a group of chemicals obtained from Indian hemp plant *Cannabis sativa* (vernacular name: **Ganjai mokka**). They interact with cannabinoid receptors present in the brain. The flower tops, leaves and the resin of this plant are used in various combinations to produce marijuana, hashish, charas and ganja. These days, cannabinoids are being abused by even some sports-persons (doping).

Mode of abuse: These are generally taken by **inhalation** and **oral ingestion**.

Effect: Show their effects on cardiovascular system of the body.

- 3. Coca alkaloid or Cocaine:** It is a white, crystalline alkaloid that is obtained from the leaves of Coca plant *Erythroxylum coca*, native to South America. It is commonly called '**coke** or **crack**'.

Mode of abuse: It is usually snorted

Effect: It has a potent stimulating action on the central nervous system as it interferes with the transport of the neuro-transmitter '**dopamine**'. Hence it produces a sense of euphoria and increased energy. (NCERT Text) Excessive consumption of crack causes hallucinations.

- 4.** Other well-known plants with hallucinogenic properties are *Atropa belladonna* and *Datura*. Certain drugs like 'Barbiturates (sleeping pills), Amphetamines (cause sleeplessness), Benzodiazepines (tranquilizers), Lysergic acid diethyl amides (LSD) and other similar drugs, normally used as medicines to treat patients with mental illnesses like depression, insomnia, etc.,' are often abused.

6.4.3 Adolescence and TDA abuse

1. Adolescence: It is the time period between the beginning of puberty and the beginning of adulthood. In other words, it is the bridge linking childhood and adulthood. The age between 12-18 years is considered '**adolescence period**'. It is both 'a period and a process' during which a child becomes mature. It is accompanied by several biological and behavioural changes. Thus, adolescence is a very '**vulnerable phase**' of mental and psychological development of an individual.

2. TDA abuse: Curiosity, desire for adventure and excitement, experimentation, are the common causes for the motivation of youngsters towards the use of tobacco, drugs and alcohol. The first use of drugs or alcohol may be out of curiosity or experimentation, but later the person starts using them to escape facing problems. Recently 'stress from the pressure to excel in academics or examinations' has played a significant role in alluring the youngsters to try certain drugs. Television, movies, newspapers and internet also help promoting this wrong perception. Other factors that are associated with tobacco, drug and alcohol abuse among adolescents are unstable or unsupportive family structures and peer pressure.

The consumption of drugs in any form is a non bailable offence. But the seeds of *Opium* are easily available in any provision store. They are treated in such a way that they can never germinate. They are generally used in the preparation of some curries. What are they? **Clue:** Find the Telugu name of poppy seeds.

6.4.4 Addiction and Dependence

The TDA abuse leads to addiction and dependence.

1. Addiction: It is a psychological attachment to certain effects such as euphoria. The most important thing one fails to realise is, the inherent '**addictive nature**' of tobacco, drugs and alcohol. With the repeated use of TDA, the tolerance level of the receptors present in our body increases. Consequently the receptors respond only to higher doses leading to greater intake and addiction. However it should be clearly borne in mind that **use of TDA even once, can be a fore-runner to addiction**. Thus, the addictive potential of tobacco, drugs and alcohol pull the users into a vicious circle leading to their regular use (abuse) from which they may not be able to get out. In the absence of any guidance or counseling, people get addicted and become dependent on them.

2. Dependence: It is the tendency of the body to manifest a characteristic and unpleasant condition (withdrawal syndrome) if the regular dose of drugs or alcohol is abruptly discontinued. The withdrawal syndrome is characterised by anxiety, shakiness (tremors), nausea and sweating which may be relieved when the regular use is resumed again. Dependence leads the patients to ignore all social norms.

6.4.5 Adverse effects of drugs and alcohol abuse

The immediate adverse effects of drugs and alcohol abuse are manifested in the form of reluctant behaviour, vandalism and violence. Excessive doses of drugs may lead to coma and death due to respiratory or heart failure or cerebral haemorrhage. A combination of drugs or their intake along with alcohol generally results in overdosing and even death. **The most common warning signs of drug and alcohol abuse among the youth include ‘drop in academic performance, lack of interest in personal hygiene, depression, fatigue, aggressive behaviour, loss of interest in hobbies, change in sleeping and eating habits, fluctuations in weight, appetite, etc’.**

Those who take drugs intravenously are much more likely to acquire serious infections such as HIV, HBV (Hepatitis-B virus), etc., as the viruses are transferred from one person to another by the sharing of infected needles and syringes. The chronic use of drugs and alcohol damages nervous system and liver. The use of drugs and alcohol during pregnancy is also known to affect the foetus adversely.

Some sports-persons take drugs such as anabolic steroids to enhance their performance. The side-effects of the use of these drugs in females include masculinisation, increased aggressiveness, mood swings, depression, abnormal menstrual cycles, excessive hair growth on the face and body and the enlargement of clitoris. In males it includes acne (pimples), increased aggressiveness, mood swings, depression, reduction in the size of testicles, decreased sperm production, kidney and liver dysfunction, enlargement of breasts, premature baldness and the enlargement of the prostate gland.

6.4.6 Prevention and Control

The age-old adage of **‘Prevention is better than cure’** holds true here also. Some of the measures useful for prevention and control of TDA abuse among the adolescents are:

- i) **Avoid undue parental pressure:** Every child has his/her own choice, capacity and personality. The parents should not force their children to perform beyond their capacity by comparing them with others in studies, games, etc.
- ii) **Responsibility of parents and teachers:** They should look for the danger signs and counsel such students who are likely to get into the ‘trap’.
- iii) **Seeking help from peers:** If peers find some one abusing drugs or alcohol, immediately it should be brought to the notice of their parents or teachers so that they can guide them appropriately.
- iv) **Education and counselling:** Educating and counselling the children to face problems, stress and failures as a part of life.
- v) **Seeking professional and medical help:** A lot of help is available in the form of highly qualified psychologists, psychiatrists and de-addiction and rehabilitation programmers.

GLOSSARY

Abscess: A wound consisting of a localized collection of pus surrounded by inflamed tissue

Abuse: Improper or excessive use.

Alkaloid: Natural bases containing nitrogen, found in plants. e.g. Quinine

Alveoli of lungs: Tiny sacs for holding air in the lungs in which the exchange of gases takes place

Anabolic steroids: Technically known as anabolic-androgen steroids (AAS) or colloquially as 'steroids', are the drugs that mimic the effects of 'testosterone and dihydrotestosterone in the body'. They increase protein synthesis within cells, which results in the buildup of cellular tissue (anabolism), especially in muscles.

Anaemia: Deficiency of red blood cells or haemoglobin

Appetite: A feeling of hunger

Bout: An instance of something lasting for a short period as in bout of fever

Clinical symptoms: Noticeable symptoms of a disease

Cloaca: The cavity at the end of the digestive tract into which the intestinal, genital, and urinary tracts open in the case of vertebrates (only intestinal and genital ducts open in the case of invertebrates)

Constipation: Irregular and infrequent or difficult egestion

Coprophagous: Feeding on faecal matter

Cosmopolitan: Occurring in many parts of the world

Emaciation: Extreme leanness

Epidemic: Spreading of an infectious disease, affecting many people over a wide area

Epidemiology: The branch of medical science dealing with the transmission and control of disease

Euphoria: A temporary feeling of an exaggerated joy, well-being, pride and optimism associated with drugs and alcohol; or an absence of depression

Faeces: Solid indigested product discharged through anus

Filiform: Resembling a thread

Folk-medicine: Treatment given by some aged local people (naatu vaidyam in Telugu)

Giddiness: A feeling that you are about to fall; a reeling sensation

Groin: The junction of the inner part of the thigh with the trunk together with the adjacent region and often including the external genitals

Hallucinations: Illusionary perception; a sensory experience of something that does not exist outside the mind, caused by physical and mental disorders

Hoarseness: Harshness of the throat

Inflammation: A response of body tissues to injury or irritation, characterized by pain, swelling, redness and burning sensation

Instinct: Inborn pattern of behaviour

Intravenous: Direct injection into the vein using a needle and syringe

Lesion: An injury to living tissue

Liver sinusoids: Tiny blood filled spaces in the liver

Masculinisation: Development of features like those of males

Moultings: Periodic shedding of the cuticle or the outer skin in some animals or larval stages

Nasal congestion: Blockage of nose

Neuro-transmitter: A neuro-chemical that transmits nerve impulses across a synapse

Ovoviviparous: Producing eggs that hatch within the body but the juveniles do not draw any nourishment from the mother

Pathogen: Any disease-producing agent

Peer: A person who is of equal standing with another in a group; a friend or a classmate

Phasmids: Caudal sensory organs present in nematodes

Pronucleus: Nuclei of the sperm and ovum before their union to form synkaryon

Quagmire: A soft wet area of low-lying land that sinks underfoot (quag means 'oobi/ oobi nela' in Telugu)

Rituals: The prescribed procedure for conducting religious ceremonies

Sausage shaped: More or less a long-brinjal shaped

Schizogony: A type of multiple fission that occurs in many apicomplexan protozoans during asexual cycle

Snorting: An act of forcible inhalation of drugs

Splenomegaly: An abnormal enlargement of the spleen

Stool: Faecal matter

Synkaryon: Nucleus of the zygote

Thrive: Live/grow vigorously

Tranquilizers: A drug used to reduce stress or tension without reducing mental clarity

Tropical sprue: A chronic disorder that occurs in both children and adults; nutrients are not properly absorbed; symptoms include foul-smelling diarrhoea and emaciation

Ulcer: A circumscribed inflammatory lesion on the skin or an internal mucous membrane resulting in the necrosis of tissue

Vaccine: Immunogen consisting of a suspension of weakened or dead pathogenic cells injected in order to stimulate the production of antibodies in the host body

Vandalism: Deliberate destruction, defacement or damage of public or other people's property

Vermiform: Long, thin and cylindrical or resembling a worm

Vesicular nucleus: Nucleus with a nucleolus and large amounts of euchromatin

Vicious circle: One trouble leads to another that aggravates the first

Vulnerable: Susceptible to attack or temptation

Welfare: Something that aids or promotes

SOME MORE COMMON DISEASES

i) **Bronchitis:** Inflammation of the membranes lining the bronchial tubes

ii) **Chikungunya:** It is a viral disease caused by *Chikungunya virus* (CHIKV) which is a heat sensitive RNA virus
Mode of infection: Inoculation through the bite of *Aedes aegypti* which bites during the day time.

Symptoms: Fever upto 104°F, chills, vomiting, nausea, headache and **severe joint pains** and sometimes it is accompanied with rashes

iii) **Diphtheria:** It is caused by a gram-positive bacillus bacterium *Corynebacterium diphtheriae*. It primarily affects the mucous membranes of the respiratory tract.

Mode of infection: Contamination through airborne respiratory droplets or by direct contact with nasopharyngeal secretions

Symptoms: Sore throat, fever, malaise, hoarseness, difficulty in swallowing or in breathing

iv) **Dysentery:** It is an intestinal inflammation, especially in the colon, caused by *Shigella*, a gram negative bacterium

Mode of infection: Contamination through food & water

Symptoms: Abdominal pain, nausea, vomiting, watery diarrhea which can contain blood, mucus or pus, painful egestion and dehydration

- v) **Emphysema:** an abnormal condition of the lungs marked by decreased respiratory function due to the rupture of alveoli

- vi) **Plague:** Bubonic plague is the most common form of plague. It is caused by a Gram-negative bacterium *Yersinia pestis*

Mode of infection: Inoculation through the bites of an infected flea (*Xenopsylla cheopis*)

Symptoms: Swollen, tender lymph glands, Fever, Chills, Headache and weakness

- vii) **Polio:** It is caused by *Poliomyelitis* virus which is a single-stranded RNA virus

Mode of infection: Contamination through the stool of the infected person.

Symptoms: Inflammation of nerve cells of the brain stem and spinal cord resulting in the impairment of legs.

- viii) **Smallpox:** It is caused by the *Variola* virus.

Mode of infection: Contamination through the tiny drops of an infected person's saliva that comes when the patient talks, coughs or sneezes.

Symptoms: Formation of small pus-filled blisters through out the body, fever, headache, backache, and feeling tired.

- ix) **Tetanus:** It is an infectious disease caused by contamination of wounds from a Gram-positive bacterium *Clostridium tetani* or its spores present in the soil.

Mode of infection: Contamination through soil or dust.

Symptoms: Tetanus results in severe, uncontrollable muscle spasms as in lockjaw.

Muscle spasm: a painful and involuntary muscular contraction.

VERY SHORT ANSWER TYPE QUESTIONS

1. Define parasitism and justify this term.
2. Distinguish between vector and a reservoir host.
3. Distinguish between mechanical vector and biological vector.
4. What is a hyper-parasite? Mention the name of one hyper-parasite.
5. What do you mean by parasitic castration? Give one example.
6. What are the endo-parasitic adaptations observed in *Fasciola hepatica*?
7. Define neoplasia. Give one example.
8. Define the most accurate definition of the term 'health' and write any two factors that affect the health.
9. Distinguish between infectious and non-infectious diseases. Give two examples each.
10. '*Entamoeba histolytica* is an obligatory anaerobe'. Justify.
11. Distinguish between precystic stage and cystic stage of *E. histolytica*.
12. What is the reserve food in the precystic and early cyst stages of *Entamoeba histolytica*?
13. A person is suffering from bowel irregularity, abdominal pain, blood and mucus in stool, etc. Based on these symptoms, name the disease and its causative organism.
14. On the advice of a doctor, a patient has gone to a clinical laboratory for the examination of a sample of faeces. The lab technician, on observing the stool of the patient diagnosed that the patient was suffering from amoebiasis. Write any two

characteristic features based on which the technician came to that conclusion.

15. Define 'asymptomatic cyst passers' with reference to *Entamoeba histolytica*.
16. What are the stages of *Plasmodium vivax* that infect the hepatocytes of man?
17. Define prepatent period. What is its duration in the life cycle of *Plasmodium vivax*?
18. Define incubation period. What is its duration in the life cycle of *Plasmodium vivax*?
19. What are Schuffner's dots? What is their significance?
20. What are haemozoin granules? What is their significance?
21. What is exflagellation and what are the resultant products called?
22. Why is the syngamy found in *Plasmodium* called anisogamy?
23. What is ookinete? Based on the 'sets of chromosomes' how do you describe it?
24. A person is suffering from chills and shivering and high temperature. These symptoms are cyclically followed by profuse sweating and return to normal body temperature. Based on these symptoms, name the disease and its causative organism.
25. Describe the methods of biological control of mosquitoes.
26. The eggs of *Ascaris* are called 'mammillated eggs'. Justify.
27. What is meant by nocturnal periodicity with reference to the life history of a nematode parasite you have studied?

28. Distinguish between lymphadenitis and lymphangitis?
29. 'Elephantiasis is the terminal condition of filariasis'. Justify.
30. In which way does tobacco affect the respiration? Name the alkaloid found in tobacco.
31. Define drug abuse.
32. From which substances 'Smack' and 'Coke' are obtained?
33. 'Many secondary metabolites of plants have medicinal properties. It is their misuse that creates problems'. Justify the statement with an example.
34. Why are cannabinoids and anabolic steroids banned in sports and games?
35. Mention the names of any four drugs which are used as medicines to treat patients with mental illness like depression, insomnia, etc.' that are often abused

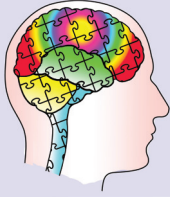
SHORT ANSWER TYPE QUESTIONS

1. What is the need for parasites to develop special adaptations? Mention some special adaptations developed by the parasites.
2. Distinguish between hypertrophy and hyperplasia with an example for each.
3. Describe the structure of a trophozoite of *Entamoeba histolytica*.
4. Explain the life cycle of *Entamoeba histolytica*.
5. Write a short note on the pathogenicity of *Entamoeba histolytica*.
6. Describe the structure of sporozoite of *Plasmodium vivax*.
7. Describe the cycle of Golgi in the life history of *Plasmodium vivax*.
8. Explain the pathogenicity of *Wuchereria bancrofti* in man.

9. Write short notes on typhoid fever and its prophylaxis.
10. Write short notes on pneumonia and its prophylaxis.
11. Write short notes on common cold and its prophylaxis.
12. Write short notes on ringworm and its prophylaxis.
13. What are the adverse effects of tobacco?
14. Write short notes on opioids.
15. Write short notes on cannabinoids.
16. Write short notes on cocaine.
17. Why in adolescence is considered vulnerable phase?
18. Distinguish between addiction and dependence.
19. 'Prevention is better than cure'. Justify with regard to TDA abuse.

LONG ANSWER TYPE QUESTIONS

1. Explain the structure and life cycle of *Entamoeba histolytica* with the help of neat and labelled diagrams.
2. Describe the life cycle of *Plasmodium vivax* in man.
3. Describe the life cycle of *Plasmodium vivax* in mosquito.
4. Describe the life cycle of *Ascaris lumbricoides* with the help of a neat and labelled diagram.
5. Describe the life cycle of *Wuchereria bancrofti*.



For ignited minds

- NOT FOR EVALUATION

1. Why should *Anopheles* only be the vector for *Plasmodium* when there are several other types of mosquitoes in the world ?
2. The epithelium surrounding the wall of the gut is called peritoneum. Inflammation of peritoneum is called peritonitis. From the information provided above, which of the following can possibly cause peritonitis in man? (List: *Plasmodium* , *Wuchereria*, *Entamoeba* and *Ascaris*).
3. If there is swelling in both the legs, it is probably a problem related to heart or kidneys. If there is a swelling in only one leg what should a doctor suspect, in your opinion.
4. Can housefly be both mechanical and biological carrier of pathogens?
5. Hypothetically if the sausage shaped larva (First stage larva) of *Wuchereria bancrofti* is introduced into blood of man , what can be expected as further consequence?
6. Probably Ronald Ross could have missed his Nobel Prize ,had there not been a scientist by the name Patrick

Manson was not born before Ross. If your answer is 'probably, yes' how do you offer an explanation in favour of your view?

7. If human stool has 'live trophozoites' of a parasite, in a sample of stool under examination, how can you identify them as those of *Entamoeba histolytica*? (if you are provided with the right stains and microscope required)
8. In the case of a suspected filarial patient why is the patient admitted in the hospital and a sample of blood is taken from him around midnight.
9. If all the female *Culex* mosquitoes in the world are hypothetically killed will there be filariasis in the world, any more ?
10. If the salivary glands of all female mosquitoes are hypothetically removed, what important effects can be expected? Can you think of at least two important effects?
11. Can you guess the expansion of 'COLD'?

Common obstructive lung disease.